① What is Engineering Economics? Explain its importance in engineering decision making??

⇒ Engineering Economics is the application of economic principles and mathematical methods to evaluate the costs, benefits and financial consequences of engineering projects.

Importance:

Cost Estimation: Helps estimate the total cost of a project.

Resource Allocation: Ensures efficient use of money materials and labour.

Alternative Selection: Helps choose the most economical option among alternatives.

Investment Decision: Helps determine whether a project is financially worthwhile.

Profit Maximization: Helps determine whether increase profit and reduces unnecessary expenses.

Risk analysis: Helps evaluate, financial risks and uncertainties.

Therefore, Engineering economics helps engineers make rational, cost effective and financially sound decisions.